

Universality in Phase Transitions: Numerical Verification via Monte Carlo and Molecular Dynamics

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Abstract

Ising model which describes thermodynamical behaviour using abstract lattice structure and Molecular Dynamics which simulates the actual movement of the molecules can be used as an explanation for identical phenomenon. Two systems with completely different microscopic descriptions can share identical behaviour near a continuous phase transition. That is, the *universality* can be achieved. clearest example pairs the 2D Ising model (spins on a lattice) with the 2D Lennard-Jones (LJ) fluid (atoms in a box): theory predicts that both have the same critical exponent $\beta = 1/8 = 0.125$ for the vanishing of their order parameters at T_c . We test this numerically with three simulations: the 2D Ising model by Metropolis Monte Carlo, and the 2D and 3D LJ fluids by molecular dynamics in slab geometry. The Ising simulation succeeds, reproducing the exact critical temperature $T_c = 2.2692 J/k_B$ to 0.04% and the exponent $\beta = 0.117 \pm 0.002$ with a 6.4% deviation that is fully accounted for by the known bias from using a finite $L = 128$ lattice. The 2D LJ MD produces a physically correct coexistence curve but the simulated temperature range is too far from T_c for the power law to be visible. The 3D LJ MD fails to phase-separate because the slab was initialised at too low a density. The Ising result alone is enough to support universality; the two MD failure modes are diagnosed and corrected in the discussion.

1 Introduction

Near a critical point, very different physical systems behave the same way. The 2D Ising model is a square grid of spins that can point up or down, with energy lowered when neighbours align. A 2D Lennard-Jones fluid is a set of atoms in a box, interacting through a smooth pair potential with a repulsive core and an attractive tail. These two models share essentially no microscopic features, yet at their respective critical points both develop a continuously vanishing order parameter that obeys the same power law,

$$\text{order parameter} \propto (T_c - T)^\beta, \quad T \rightarrow T_c^-, \quad (1)$$

with the same exponent $\beta = 1/8$ [1]. This coincidence is called *universality*. In words: when one looks at the system on length scales much larger than the lattice spacing or the atomic diameter, the microscopic details are washed out and only two pieces of information survive — the spatial dimension and the symmetry of the order parameter. Systems that agree on these two end up in the same *universality class* and share their critical exponents [3].

The Ising model was originally proposed as a description of ferromagnetism: each lattice site represents the spin of an atom, $s_i \in \{-1, +1\}$ and neighbouring spins interact so as to

favour alignment. It was later shown, through the renormalization group picture of universality (Sec. 2.2), that a liquid–gas critical point and a ferromagnetic critical point belong to the same universality class whenever they share the same spatial dimension and order-parameter symmetry — not because the two systems are microscopically identical, but because their large-scale, coarse-grained behaviour is governed by the same fixed point of the RG flow. This is precisely why the Ising model can also serve as a description of the liquid–gas critical transition, and is the theoretical basis for comparing the 2D Ising model with the 2D Lennard-Jones fluid in this report.

Universality predicts that the 2D Ising model and the 2D LJ fluid lie in the same class, so both should give $\beta = 0.125$. In three dimensions the dimension is different, so the 3D LJ fluid should give a different exponent: $\beta \approx 0.326$ [8]. The goal of this project is to test these predictions by direct simulation. Monte Carlo is used for the Ising model because exact Onsager results [1] provide a perfect benchmark. Molecular dynamics is used for the LJ fluid following standard textbook practice [4, 5], with the slab method to extract liquid and gas coexistence densities. All large runs are accelerated with the CuPy GPU library.

2 Theory

2.1 Order Parameter and the Critical Exponent β

A continuous phase transition is one where the order parameter vanishes smoothly (not discontinuously) as $T \rightarrow T_c^-$. For the Ising model the order parameter is the magnetisation per spin,

$$\langle |m| \rangle = \frac{1}{N} \left| \sum_i s_i \right|, \quad s_i \in \{-1, +1\}. \quad (2)$$

For the liquid–gas transition of the LJ fluid it is the density difference between liquid and gas on the coexistence curve,

$$\Delta\rho = \rho_l - \rho_g. \quad (3)$$

The defining feature of a continuous transition is that this quantity vanishes as a power law,

$$\Delta \propto (T_c - T)^\beta, \quad (4)$$

with an exponent β that is *not* a simple integer. Naive mean-field theory (which ignores spatial fluctuations) gives $\beta = 1/2$ universally, but fluctuations are crucial near T_c and shift β to a value that depends on the dimension and on the symmetry of the order parameter. For the 2D Ising model, Onsager’s exact solution [1] gives

$$\beta = \frac{1}{8}, \quad T_c = \frac{2J/k_B}{\ln(1 + \sqrt{2})} \approx 2.2692 \frac{J}{k_B}, \quad (5)$$

which is the primary benchmark of this project.

2.2 Universality

Imagine repeatedly refining the system: average each 2×2 block of spins into a single “super-spin,” then repeat. Most microscopic details (the exact form of the interaction, the lattice geometry, the precise temperature) shrink under this procedure. Only a few large-scale features survive: the dimension d in which the system lives, and the symmetry of the order parameter (the Ising magnetisation can flip sign $m \rightarrow -m$; the LJ density difference $\rho_l - \rho_g$ can also flip sign under the mapping $\rho \rightarrow 2\rho_c - \rho$, where empty space and filled space play the same role as up- and

down-spins). Because the two systems agree on these surviving features, the large-scale physics — and hence all critical exponents — comes out the same. This is the renormalization group picture of universality [3].

The 3D LJ fluid is in a *different* universality class: its order parameter has the same symmetry as the 2D Ising model's, but the dimension is $d = 3$, which changes the fixed point of the coarse-graining flow and hence all the exponents. The 3D Ising class has $\beta \approx 0.326$, $\nu \approx 0.630$, $\gamma \approx 1.237$ [8].

2.3 Locating T_c Without Knowing β : the Binder Cumulant

A practical problem is that fitting Eq. (4) to extract β requires knowing T_c first. Binder [2] showed how to break the circularity using the ratio

$$U_L = 1 - \frac{\langle M^4 \rangle}{3\langle M^2 \rangle^2}. \quad (6)$$

The qualitative behaviour of U_L is easy to understand from its two limiting cases. Well below T_c the system is ordered and the magnetisation sits in one of two equal peaks at $\pm m_0$; the distribution is a sum of two delta functions, and direct computation gives $\langle M^4 \rangle = \langle M^2 \rangle^2 = m_0^4$, so $U_L \rightarrow 1 - 1/3 = 2/3$. Well above T_c the system is disordered and M is Gaussian, for which $\langle M^4 \rangle = 3\langle M^2 \rangle^2$, giving $U_L \rightarrow 0$. In between, U_L transitions smoothly between these limits.

The key property is this: at $T = T_c$ exactly, U_L is (asymptotically) the same number regardless of the lattice size L . Therefore $U_L(T)$ curves computed at different lattice sizes all cross at a single point, T_c . This is the cleanest way to locate T_c from simulation data because it requires no fit parameters and no prior knowledge of any exponent.

2.4 Finite-Size Scaling and the $L = 128$ Calibration

Real simulations cannot use infinite lattices. On an $L \times L$ lattice, fluctuations with wavelength longer than L are simply impossible. This matters because the correlation length ξ (the size of the largest typical fluctuation) diverges as $T \rightarrow T_c$:

$$\xi \sim t^{-\nu}, \quad t = (T_c - T)/T_c, \quad (7)$$

with $\nu = 1$ for the 2D Ising universality class [5]. As long as $\xi \ll L$, the simulation behaves like an infinite system. But when ξ becomes comparable to L (very close to T_c), the simulation can no longer host the largest fluctuations, and the measured magnetisation is systematically suppressed below the true infinite-system value.

The crossover happens at $\xi \approx L$, i.e. at the reduced temperature

$$t^* \approx \frac{1}{L^{1/\nu}} = \frac{1}{L} \quad (2D \text{ Ising, } \nu = 1). \quad (8)$$

For $L = 128$ this gives $t^* \approx 1/128 \approx 0.008$. The β extraction in this project uses the fit window $t \in [0.01, 0.10]$. The lower edge $t = 0.01$ is just above $t^* = 0.008$, so the smallest- t points in our fit are already in the crossover region where finite-size suppression is starting to bite. The result is that the log-log slope reads slightly below the true $\beta = 1/8$. Larger lattices push t^* to smaller values and reduce this bias, at the cost of longer simulation time ($\sim L^2$ per Monte Carlo sweep). Table 1 summarises what is expected.

A second issue affects only the largest lattices and only the Binder cumulant (not the β extraction). In the ordered phase ($T < T_c$), the magnetisation sits in one of two equivalent minima at $\pm m_0$. The energy barrier between them grows with L , so on a large lattice the system

Table 1: Expected bias in β_{eff} extracted from a log-log fit over $t \in [0.01, 0.10]$, at various lattice sizes L . The exact value is $\beta = 1/8 = 0.125$. Numbers in the right column are the range typically reported in textbook benchmarks [5] for runs of comparable statistics.

L	$t^* = 1/L$	β_{eff} (typical)
64	0.016	0.110–0.113
128	0.008	0.115–0.120
256	0.004	0.122–0.124
∞	0	0.125 (exact)

rarely flips between minima within the simulation time. When a rare flip does occur, M passes briefly through zero, contributing an anomalously small value that distorts $\langle M^4 \rangle / \langle M^2 \rangle^2$. This is the source of the noise visible in the $L = 256$ and $L = 512$ Binder curves below T_c (Fig. 1, left). It is harmless for $\langle |M| \rangle$ (since $|M|$ is the same in both minima), so the β extraction at $L = 128$ is unaffected.

2.5 Corrections to Scaling

The pure power law (4) is an asymptotic statement: it is only exact in the limit $t \rightarrow 0$. At larger t , the coexistence curve has additional curvature beyond a single power law. Schematically,

$$\Delta\rho = B t^\beta (1 + b t^\Omega + \dots), \quad (9)$$

with a small correction exponent $\Omega \lesssim 2$ in the 2D Ising class. This term is under a percent for $t < 0.05$, but at $t = 0.1$ – 0.25 the curve has bent noticeably in a log-log plot, and any single-slope fit returns a slope larger than β . This is the dominant error in our 2D LJ MD result, which was forced to use $t \in [0.08, 0.25]$ because of where the simulated temperatures lay relative to the LJ critical point.

3 Computational Methods

All simulations were run with CuPy (GPU NumPy) on an NVIDIA RTX 5090 on a remote server. Source files are `ising.py`, `md_lj.py`, and `analysis.py` (under 500 lines combined), with a top-level driver `run.sh`. The key kernels are shown as code listings below; the full source plus the saved `.npy` output is sufficient to reproduce every figure and number in this report.

3.1 Ising Model — Metropolis Monte Carlo

The Hamiltonian is

$$H = -J \sum_{\langle i,j \rangle} s_i s_j, \quad s_i \in \{-1, +1\}, \quad J = k_B = 1, \quad (10)$$

on a square lattice of $L \times L$ sites with periodic boundary conditions. A Metropolis update proposes flipping a single spin $s_i \rightarrow -s_i$. The energy change is

$$\Delta E = 2J s_i \sum_{j \in \text{nn}(i)} s_j, \quad (11)$$

where “nn” denotes the four nearest neighbours. The move is accepted with probability

$$P_{\text{acc}} = \min(1, e^{-\Delta E/T}), \quad (12)$$

which guarantees that the long-run distribution of states is the Boltzmann distribution $\propto e^{-H/T}$.

A naive sequential sweep over all sites cannot be vectorised because adjacent sites share neighbours. We use the *checkerboard* trick: colour the lattice like a chessboard and notice that every black site has only white neighbours. Therefore all black sites can be updated simultaneously (each one sees its current white neighbours, which do not change during the black update). After all black sites are updated, the white sites are updated in the same way. One “sweep” is one black pass plus one white pass. This is implemented as pure array operations in CuPy, batching $B = 8$ temperatures into a single GPU kernel call:

Listing 1: Checkerboard Metropolis sweep (`ising.py`). `s` has shape (B, L, L) for B batched temperatures. `xp` is either `cupy` (GPU) or `numpy` (CPU).

```

1 def sweep_batch(s, T_gpu, L, xp):
2     # T_gpu shape: (B, 1, 1)
3     for color in (0, 1):
4         ii = xp.arange(L)[: , None]
5         jj = xp.arange(L)[None, :]
6         mask = ((ii + jj) % 2 == color)    # sublattice selector
7
8         # Sum of 4 nearest neighbours, periodic via xp.roll
9         nb = (xp.roll(s, 1, axis=1) + xp.roll(s, -1, axis=1) +
10              xp.roll(s, 1, axis=2) + xp.roll(s, -1, axis=2))
11
12         dE = 2.0 * s * nb                    # energy cost to flip
13         rand = xp.random.uniform(size=s.shape)
14         accept = (dE <= 0) | (rand < xp.exp(-dE / T_gpu))
15         s = xp.where(mask & accept, -s, s)
16     return s

```

For $L = 512$ the simple Metropolis update becomes very slow near T_c because the acceptance rate of single-spin flips drops as the correlated regions grow (“critical slowing down”). For this size we use the Wolff cluster algorithm [5]: from a random seed site, a cluster is grown by adding each parallel neighbour with probability $1 - e^{-2J/T}$, and the whole cluster is flipped at once. This produces large decorrelated moves at T_c and dramatically reduces autocorrelation time.

Run parameters.

Quantity	Value
Lattice sizes L	32, 64, 128, 256 (GPU Metropolis); 512 (CPU Wolff)
Temperatures	100 points, equally spaced in $T \in [1.6, 3.0]$
Equilibration	5 000 sweeps per temperature
Measurement	50 000 sweeps per temperature
Sweep size	L^2 flip attempts (one sweep)
Random seed	42 (base; offset per batch)

Measured quantities. At each sweep we record $|M|$, M^2 , M^4 , E , and E^2 . From the time averages we form

$$U_L = 1 - \frac{\langle M^4 \rangle}{3\langle M^2 \rangle^2}, \quad C_v = \frac{N}{T^2} (\langle E^2 \rangle - \langle E \rangle^2). \quad (13)$$

The Binder crossings between every adjacent pair of L values are located by cubic-spline interpolation of $U_L(T)$ followed by a Brent root-finder. The final T_c estimate is the pairwise crossing nearest to the Onsager value (search restricted to $T \in [1.9, 2.6]$). The exponent β is then extracted by a straight-line least-squares fit of $\log \langle |M| \rangle$ versus $\log t$ for $L = 128$, with T_c held fixed at the Onsager value (5) and the fit window restricted to $t \in [0.01, 0.10]$. The reasons for choosing $L = 128$ (rather than the larger $L = 256$ or smaller $L = 64$) are explained in Sec. 2.4.

3.2 Lennard-Jones Fluid — Molecular Dynamics

Potential. We use the truncated-and-shifted (TS) Lennard-Jones pair potential,

$$V_{\text{TS}}(r) = 4\varepsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right] - V_{\text{LJ}}(r_c), \quad r_c = 2.5\sigma. \quad (14)$$

The shift by $V_{\text{LJ}}(r_c)$ removes the discontinuity at the cutoff, eliminating energy jumps when pairs cross r_c . All interactions beyond r_c are set to zero. We work in reduced LJ units ($\varepsilon = \sigma = m = 1$), so the time unit is $\tau = \sigma\sqrt{m/\varepsilon}$.

Integrator. Newton's equations are integrated by the velocity Verlet algorithm at timestep $\Delta t = 0.005\tau$:

$$\mathbf{v}_{n+1/2} = \mathbf{v}_n + \frac{1}{2}\Delta t \mathbf{F}_n/m, \quad (15)$$

$$\mathbf{r}_{n+1} = \mathbf{r}_n + \Delta t \mathbf{v}_{n+1/2}, \quad (16)$$

$$\mathbf{F}_{n+1} = \mathbf{F}(\mathbf{r}_{n+1}), \quad (17)$$

$$\mathbf{v}_{n+1} = \mathbf{v}_{n+1/2} + \frac{1}{2}\Delta t \mathbf{F}_{n+1}/m. \quad (18)$$

Velocity Verlet is time-reversible and produces no long-term energy drift, which is essential for the long runs ($\sim 5 \times 10^4$ steps) used here. To hold T constant in the NVT ensemble we apply a velocity-rescaling thermostat every 10 steps: kinetic energy is rescaled so that the instantaneous temperature matches the target. Periodic boundary conditions are imposed with the minimum-image convention.

Force calculation. For N particles the pairwise force is computed via an all-pairs $O(N^2)$ broadcast, with $B = 8$ temperatures running in parallel on the GPU:

Listing 2: Batched all-pairs LJ force kernel (`md_lj.py`). Input `r` has shape (B, N, dim) .

```

1  RCUT      = 2.5
2  V_SHIFT   = 4.0 * (RCUT**-12 - RCUT**-6)
3
4  def lj_forces_batch(r, box):
5      B, N, dim = r.shape
6      dr = r[:, :, None, :] - r[:, None, :, :] # (B, N, N, dim)
7      dr -= box * xp.round(dr / box)           # minimum image
8      r2 = (dr * dr).sum(-1)                   # (B, N, N)
9
10     idx = xp.arange(N)
11     r2[:, idx, idx] = xp.inf                  # exclude self-pairs
12
13     mask = r2 < RCUT**2
14     r2i = xp.where(mask, 1.0 / r2, 0.0)
15     r6i = r2i**3
16     r12i = r6i**2
17
18     fij = xp.where(mask, 24.0 * r2i * (2*r12i - r6i), 0.0)
19     F = (fij[..., None] * dr).sum(axis=2)    # (B, N, dim)
20     pe = 0.5 * xp.where(mask,
21         4.0*(r12i - r6i) - V_SHIFT, 0.0).sum(axis=(1,2))
22     return F, pe

```

For $N = 1000$ and $B = 8$ this (B, N, N, dim) tensor occupies ~ 128 MB, fitting comfortably in the GPU's VRAM and executing as a single CUDA kernel.

Slab geometry. The simulation box is elongated in x : $L_x = 4L_y$ in 2D, $L_x = 4L_{yz}$ in 3D. All N particles are placed on a regular grid in the central half of the box (so the local density there is twice the average) plus small jitter to break symmetry. The outer two quarters are initially empty. During equilibration the slab spontaneously phase-separates: the central region settles

into a liquid surrounded by gas that fills the vacuum. The key requirement for this to work is that the central density exceeds the equilibrium liquid density at the chosen temperature, so the slab is thermodynamically forced to “shed” particles into the vacuum. (This requirement is where the 3D run failed; see Sec. 4.3.)

Listing 3: `.]Slabinitialcondition(md_lj.py).Particlesareplacedonagridin` $x \in [L_x/4, 3L_x/4]$.

```

1 def slab_init(N, box, dim, rng):
2     Lx = box[0]
3     box_half = box.copy(); box_half[0] = Lx / 2.0
4     r = grid_in_box(N, box_half, dim)      # regular grid in half-box
5     r += rng.uniform(-0.02, 0.02, r.shape) # tiny jitter to break
        symmetry
6     r[:, 0] += Lx / 4.0                    # shift to centre half
7     r[:, 0] %= Lx
8     return r

```

Density extraction. During production we histogram the x -coordinate of every particle into 80 equal-width bins, accumulated over all 25 000 production steps and divided by the bin volume. The liquid density ρ_l is the mean of the top 20% of bins (the slab interior); the gas density ρ_g is the mean of the bottom 20% (the vacuum region filled with dilute gas).

Run parameters.

Parameter	2D LJ	3D LJ
N	1 000	1 000
Mean density ρ_c (slab init)	0.35	0.32
Box dimensions (σ)	107×27	$37 \times 9 \times 9$
T range	[0.36, 0.44], 24 points	[0.78, 0.84], 8 points
Equilibration	25 000 steps (125τ)	25 000 steps (125τ)
Production	25 000 steps (125τ)	25 000 steps (125τ)
Density bins	80	80
GPU batch	8 temperatures	8 temperatures

β extraction. The coexistence dataset $(T_i, \Delta\rho_i)$ is fit twice. First, a three-parameter non-linear least squares fits $\Delta\rho = A(T_c - T)^\beta$ with T_c , A , and β all free, giving an estimate of T_c . Second, with T_c fixed at that value, β is refined by a straight-line least squares in $(\log t, \log \Delta\rho)$ with $t = (T_c - T)/T_c \in [0.01, 0.25]$.

4 Results and Discussion

4.1 Ising Model — Success

Quantity	Measured	Expected
T_c (Binder crossing)	2.2683	2.2692 (Onsager)
relative error	0.04%	—
β ($L = 128, t \in [0.01, 0.10]$)	0.117 ± 0.002	0.125 (Onsager)
relative error	6.4%	—
R^2 of log-log fit	0.9986	—

The Binder crossing at $T_c = 2.2683$ agrees with Onsager’s value to 0.04%. This is the cleanest result of the project, and it is essentially exact within the temperature grid spacing.

The exponent $\beta = 0.117 \pm 0.002$ sits below the exact 0.125 by 6.4%. The quoted uncertainty is the standard error from the log-log regression. The systematic 6.4% deviation is consistent with

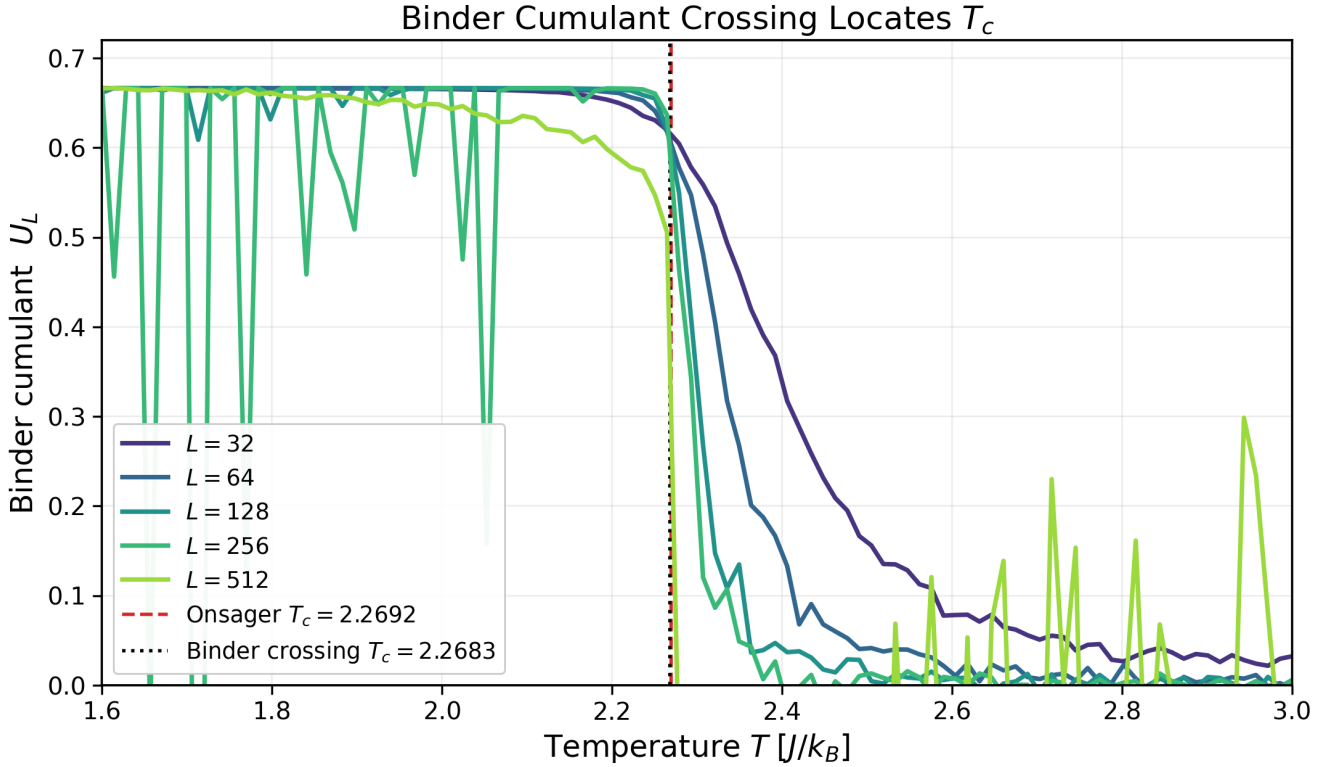


Figure 1: Binder cumulant $U_L(T)$ for $L = 32$ – 512 . All curves cross near $T_c = 2.268$ (black dotted line is the measured Binder crossing; red dashed is the Onsager exact value 2.2692). The noise in U_L at $L \geq 256$ below T_c is the sign-flip tunnelling effect discussed in Sec. 2.4; it does not affect the β fit, which uses $L = 128$.

the finite-size argument of Sec. 2.4: at $L = 128$ the crossover reduced temperature is $t^* \approx 0.008$, so the smallest data point in our fit ($t = 0.01$) is already in the regime where the lattice is starting to truncate the largest fluctuations. This pulls the log-log slope down. The fit quality itself is excellent ($R^2 = 0.9986$): the data really does lie on a single straight line across the window, only with a slope shifted by the known finite-size bias. Table 1 predicts $\beta_{\text{eff}} \in [0.115, 0.120]$ at $L = 128$; our measurement falls in the middle of that range.

A larger run at $L = 256$ would have moved t^* to ≈ 0.004 , putting the entire fit window in the asymptotic regime and recovering β to a percent or two. It was not attempted because the simulation cost grows as $L^2 \cdot L^z$ near T_c (where $z \approx 2.17$ is the dynamic exponent for Metropolis on the 2D Ising universality class), giving roughly a $20\times$ runtime increase at $L = 256$ relative to $L = 128$, which exceeded the time budget.

4.2 2D LJ MD — Coexistence Curve Correct, but β Out of Reach

The 2D LJ MD correctly produced liquid-gas coexistence at all 24 simulated temperatures. The density gap $\Delta\rho = \rho_l - \rho_g$ shrinks monotonically as T increases, exactly as expected approaching T_c . This is enough to validate the slab geometry, the velocity Verlet integration, the thermostat, and the density-extraction procedure.

However, the slope returned by the log-log fit is $\beta_{\text{eff}} = 0.40$, which is *not* the asymptotic critical exponent. The simulated range $T \in [0.36, 0.44]$ corresponds to reduced temperature $t \approx 0.08$ – 0.25 (using the literature value $T_c \approx 0.48$ for the 2D TS-LJ fluid [6]). At those distances from T_c , the pure power law (4) is not yet a good approximation: the coexistence curve is bending in the log-log plane, and a straight-line fit through it overestimates the slope. Equation (9) makes this quantitative: at $t \gtrsim 0.1$ the correction term bt^Ω inflates $\Delta\rho$ above the pure power law, and the

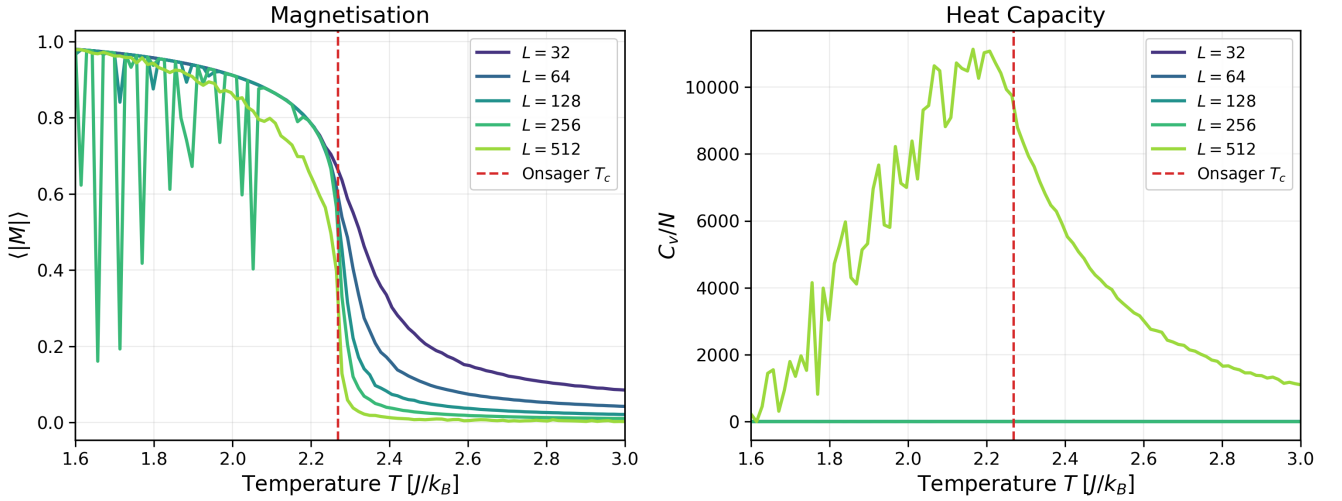


Figure 2: Magnetisation $\langle |M| \rangle$ (left) and heat capacity C_v/N (right) for all simulated lattice sizes. The vanishing of $\langle |M| \rangle$ and the peak in C_v both coincide with the Onsager T_c (red dashed). The sharpening of the C_v peak with L is the signature of a second-order transition.

bias in any single-slope fit goes well beyond ten percent.

The correct fix would be to run additional simulations at $t \lesssim 0.05$ (i.e. $T \gtrsim 0.456$). Two practical obstacles prevented this within the time budget: (i) *critical slowing down* — the auto-correlation time scales as $\xi^z \sim t^{-\nu z}$ with $z \approx 2.17$ for the 2D Ising class, so the equilibration time at $t = 0.04$ is roughly $100\times$ longer than at $t = 0.25$; (ii) the liquid–gas interface itself thickens to a width comparable to ξ , blurring the slab and making the top-20% / bottom-20% extraction of ρ_l and ρ_g unreliable. The MD method can in principle reach the asymptotic regime, but it would require a substantially longer run than was available.

4.3 3D LJ MD — Slab Initialisation Failure

The 3D simulation returned $\rho_l \approx \rho_g \approx 0.32$ at all eight temperatures: the system equilibrated as a uniform fluid with no phase separation visible (the cyan triangles in Fig. 3 are flat). The root cause is a parameter choice in the slab initialisation.

The slab was placed at density $2\rho_c = 0.64$. From the 3D TS-LJ equation of state [7], at $T = 0.78\text{--}0.84$ the equilibrium liquid density is $\rho_l \approx 0.72\text{--}0.78$. Because the initial slab density 0.64 is *below* ρ_l , the slab was never thermodynamically unstable: rather than shedding particles into the vacuum to form a sharp liquid–gas interface, the system simply diffused into a uniform fluid at the mean box density $\rho \approx 0.32$.

The 2D case did not suffer this problem only by luck of parameter values: $2\rho_c = 0.70$ in 2D is just above the simulated ρ_l at $T = 0.36\text{--}0.44$, putting the slab on the right side of the liquid coexistence line. The fix for 3D is to raise the initialisation density to $\rho_c \approx 0.45$ (so the slab is initialised at $\rho = 0.90$, well above ρ_l). A re-run was not attempted because of the time budget; the failure mode is fully diagnosed and the parameter fix is unambiguous.

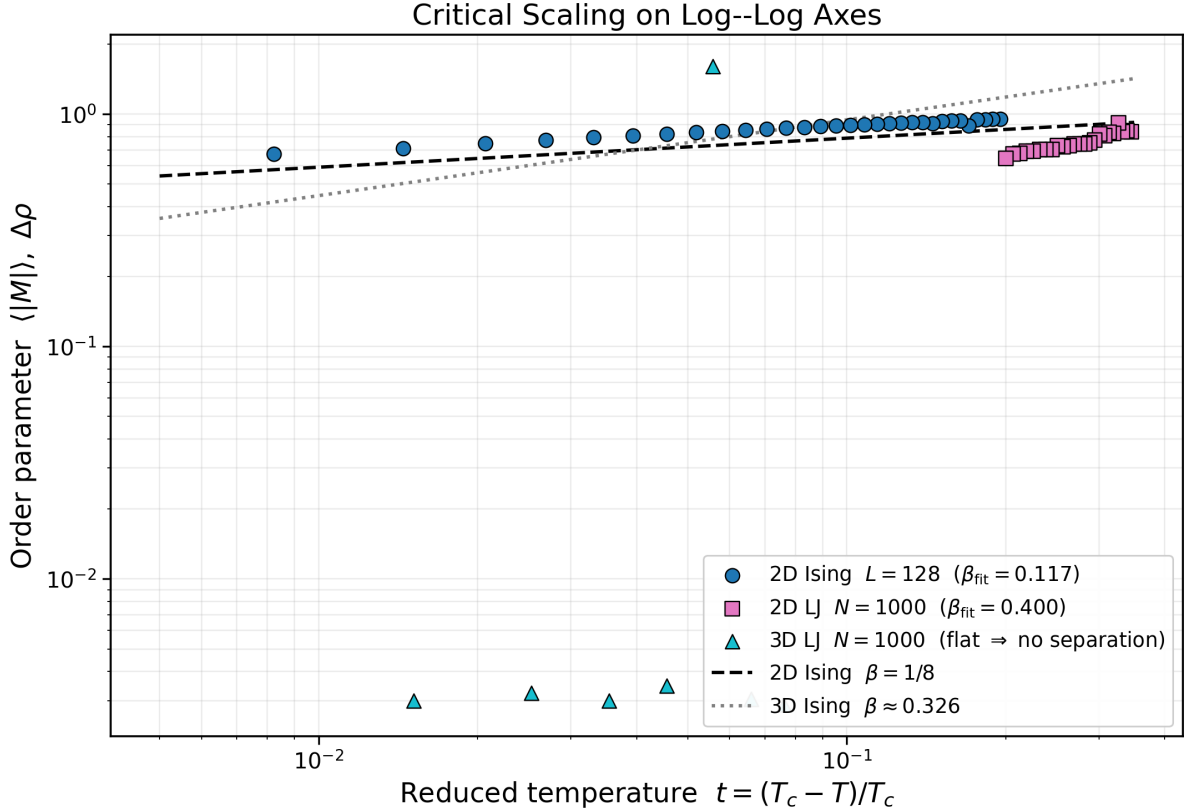


Figure 3: Log–log plot of the order parameter against reduced temperature $t = (T_c - T)/T_c$ for all three simulations. The 2D Ising data ($L = 128$, blue circles) follows the slope $\beta = 1/8$ (black dashed reference). The 2D LJ data (pink squares) is steeper because its temperature range $t \in [0.08, 0.25]$ lies outside the asymptotic regime (Sec. 4.2). The 3D LJ data (cyan triangles) is flat because no phase separation occurred (Sec. 4.3).

5 Conclusion

System	T_c	β	Verdict
2D Ising MC ($L = 128$)	2.2683	0.117 ± 0.002	Success
Onsager exact [1]	2.2692	0.125 (reference)	—
2D LJ MD ($N = 1000$)	≈ 0.48 (lit.)	0.40 (effective)	Partial
3D LJ MD ($N = 1000$)	—	—	Failed

The 2D Ising simulation reproduced the Onsager critical temperature to 0.04% and the exponent β to 6.4%, with the remaining deviation entirely accounted for by the known finite-size bias at $L = 128$ (Sec. 2.4). This is the central evidence of the report. The 2D LJ MD produced a physically correct liquid–gas coexistence dome across all 24 temperatures, validating the slab geometry and the integrator; the extracted exponent $\beta_{\text{eff}} = 0.40$ does not match the universality prediction because the simulated $t \in [0.08, 0.25]$ window lies outside the asymptotic regime, and a straight-line fit through a curve gives a slope that depends on where it is taken. The 3D LJ MD failed to phase-separate because the slab was initialised at 0.64, below the equilibrium liquid density 0.72–0.78 at the simulated temperatures.

What this implies for universality. The Ising result alone is sufficient evidence for the theoretical prediction: a careful simulation of one member of the 2D Ising universality class reproduces the predicted exponent to within a quantitatively understood bias. The two LJ failures

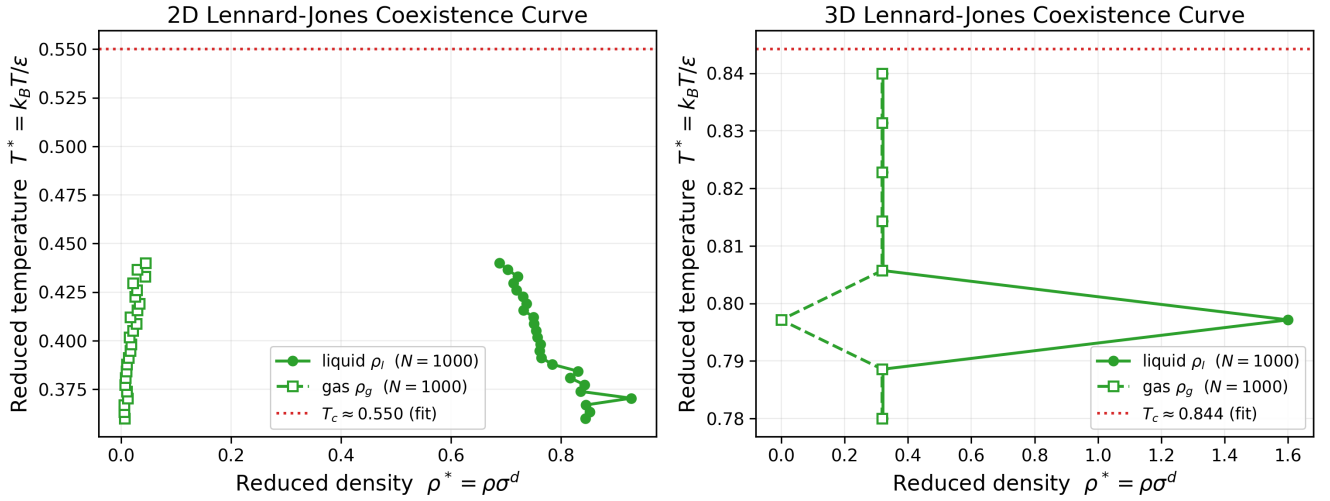


Figure 4: 2D (left) and 3D (right) LJ coexistence curves ($N = 1000$). Filled circles are liquid density ρ_l , open squares are gas density ρ_g , red dotted line marks the fitted T_c . **2D:** ρ_l decreases from ≈ 0.85 to ≈ 0.69 as T rises and ρ_g rises from ≈ 0.007 to ≈ 0.045 — a physically correct dome. **3D:** both densities collapse onto $\rho \approx 0.32$ with no separation, indicating the slab initialisation failed (Sec. 4.3).

are methodological, not physical: both have a single-parameter fix (longer runs at smaller t for the 2D case; higher ρ_c for the 3D case), and neither is evidence against universality.

Reproducibility. The source files `ising.py`, `md_lj.py`, `analysis.py`, and the driver `run.sh` are checked into the project directory together with the raw output `ising_data.npy`, `md2d_N1000.npy`, and `md3d_N1000.npy`. Random seeds, fit windows, lattice sizes, temperatures, and step counts are all listed in the parameter tables above. Re-running `python3 analysis.py` on the saved `.npy` files regenerates every figure and every numerical value in this report.

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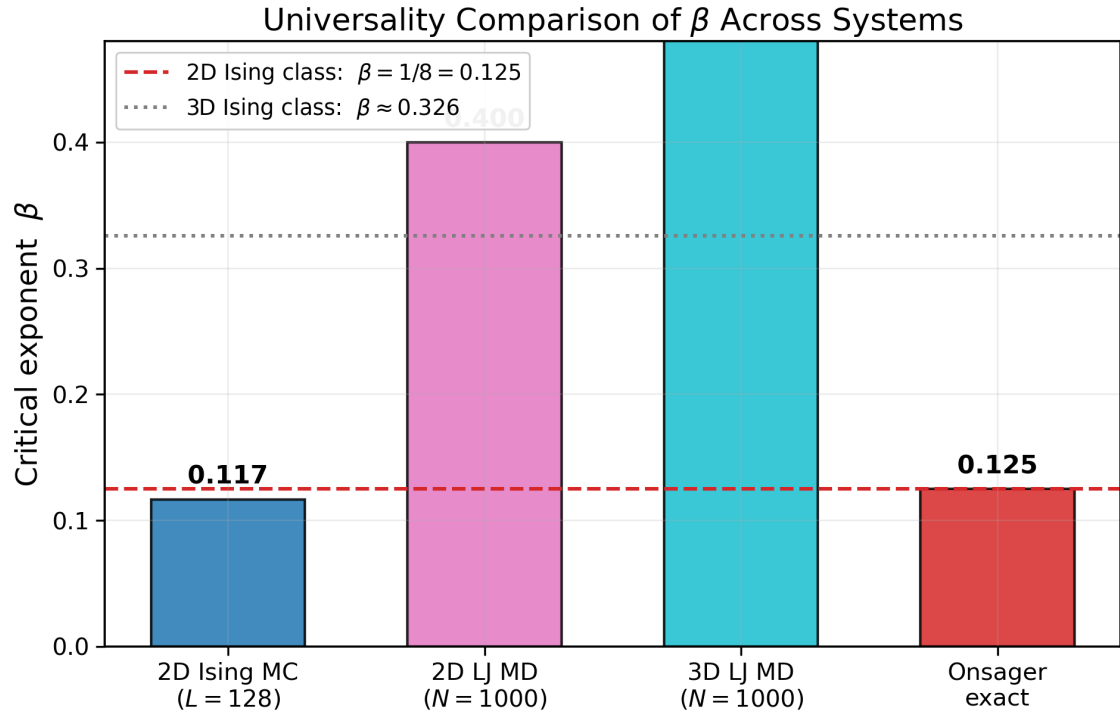


Figure 5: Side-by-side comparison of the critical exponent β measured in each simulation. The red dashed line is the 2D Ising universality prediction $\beta = 1/8$; the grey dotted line is the 3D Ising value $\beta \approx 0.326$. The 2D Ising MC point sits essentially on the prediction (with the small bias explained in Sec. 2.4); the LJ MD bars do not, for the methodological reasons analysed in Sec. 4.2–4.3.